



First Record of a Successful Nesting of South Philippine Hawk-Eagles (*Nisaetus pinskeri*) on Mindanao, Philippines

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The South Philippine Hawk-Eagle (*Nisaetus pinskeri*) is an uncommon forest raptor endemic to the Philippines (Preleuthner and Gamauf 1998, Kennedy et al. 2000, BirdLife International 2024). The species is distributed across forested areas in the western, central, and eastern Visayan Islands, including Negros, Siquijor, Bohol, Samar, Biliran, and Leyte, as well as Mindanao and Basilan in the southern Philippines (Preleuthner and Gamauf 1998, Kennedy et al. 2000).

According to the latest International Union for Conservation of Nature (IUCN) Red List assessment, the species is declining, with an estimated global population of 1300–3600 mature individuals (BirdLife International 2024). Mindanao Island is the species' stronghold (Preleuthner and Gamauf 1998), supporting an estimated 555–1580 breeding pairs. Much smaller populations persist on Negros (10–40 pairs), Bohol (5–20 pairs), and Samar/Leyte (80–175 pairs). These numbers justify the species' classification as endangered under IUCN Criterion C2a, which applies to species with small, declining populations in which no single subpopulation contains more than 1000 mature individuals (IUCN 2012). Deforestation, along with hunting and trapping, are the primary drivers of its decline, which is estimated at over 50% across the last three generations (BirdLife International 2024).

For more than a decade, our field surveys across Samar, Leyte, and Mindanao Islands have yielded repeated observations of adult, immature, or both age classes of the South Philippine Hawk-Eagle, suggesting

possible breeding activity in these areas. Complementing these field observations, online biodiversity databases also report numerous records across the species' range, with approximately 555 observations on eBird (2025) and around 674 on GBIF.org (2025). However, these online records, which also include sightings of both adults and immatures, provide only indirect evidence of breeding. Despite these numerous pieces of evidence, there remains a significant lack of ecological and biological information on the South Philippine Hawk-Eagle (Kennedy et al. 2000, Fjeldsá 2020). This knowledge gap hinders the development of targeted and effective conservation strategies. Locating an occupied nest has proven to be especially challenging, even in areas where the species is frequently observed. In this paper, we report the first documented discovery of a South Philippine Hawk-Eagle nest on Mindanao Island, Philippines, and we present preliminary observations on nesting behavior, juvenile development, and feeding ecology.

METHODS AND RESULTS

We were initially searching for a Philippine Eagle (*Pithecophaga jefferyi*) nest from 13 June to 25 August 2023. We suspected there was a nest within the Energy Development Corporation Geothermal Facility in Kidapawan City, North Cotabato Province, Mt. Apo Natural Park. Because diurnal raptors in tropical forests often rely on soaring flight (Andersen 2007), we constructed four elevated observation platforms to optimize visual coverage.

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Figure 1. Placement and structure of the South Philippine Hawk-Eagle (*Nisaetus pinskeri*) nest documented on 21 July 2023 in Mount Apo Natural Park. (A) The nestling is laying down in the nest while the adult female stands beside it, grasping a Bagobo rat (*Bullimus bagobus*); (B) the same nest showing the nestling standing while the adult female perches on a nearby branch. Photo by R. L. Taraya.

From these platforms, we heard the food-begging calls of an immature South Philippine Hawk-Eagle, and at the same time, we saw an adult South Philippine Hawk-Eagle soaring over a nearby ridge multiple times. We then observed an adult South Philippine Hawk-eagle carrying a sprig into a densely forested slope, suggesting potential nest-building activity. Investigating the area on 21 July 2023, we found a South Philippine Hawk-Eagle nest containing one nestling approximately 70 m from one of our platforms (Fig. 1). When adults visited the nest, we distinguished the male from the female by size and by distinctive plumage: the male had darker breast and facial plumage than the female. Behavioral cues further supported identification, as the female was observed feeding the young, while the male was not seen doing so, and instead appeared limited to the role of delivering prey to the nest. When we found the nest, the nestling appeared bulkier than the adults, with dense plumage. Its flight feathers were tipped with remnants of down, indicating recent emergence, and its remiges were still growing. Based on its plumage and associated behavior, we estimated the nestling to be approximately 7 wk old (Fig. 2; Philippine Eagle Center pers. comm.).

The nest was built on a tertiary forking branch of a *Terminalia* sp., atop an epiphytic shrub (*Heptapleurum simplicifolium*), situated in a riparian forest at approximately 1,200 masl within Mt. Apo Natural Park on the Balabag side of Kidapawan City, North Cotabato. The immediate nest area was relatively undisturbed with continuous canopy cover, although small-scale clearings occurred nearby. Below the nest tree and in the adjacent riparian strip, coffee and abaca had been

planted but were no longer actively farmed. As these stands became overgrown, natural trees regenerated within them. Uphill from the nest site, the landscape formed a mosaic of coffee plots, a small palm oil plantation, mixed fruit trees (e.g., durian, mangosteen), and patches of cogon grass (*Imperata cylindrica*). These clearings (<3 ha) were originally forest but had been converted to small-scale cultivation. The nearest human dwellings consisted of only three small thatched huts, and overall disturbance was minimal. Despite these land-use modifications, the riparian forest where the nest was located was part of the extensive tracts of continuous riparian and forested mountain areas of Mt. Apo Natural Park. The nesting site supported earlier assertions that the species was forest-dependent (Preleuthner and Gamauf 1998), while its proximity to small-scale clearings and regenerating farmland underscored its persistence in partially modified habitats and highlighted the need to better understand the South Philippine Hawk-Eagle's tolerance of human-modified landscapes.

We monitored the nest daily from 21 July to 25 August 2023, when weather conditions permitted, for a total of 145.7 observation hours. We monitored via a 20 × 56 field scope, 10 × 42 binoculars, and a Canon 800D camera with a 600-mm Tamron lens. Following the Philippine Eagle ethogram developed by Ibañez et al. (2003), we documented the birds' behavior with emphasis on nestling development. We recorded the nestling's daily activities and the duration of each behavior, and calculated the proportion of time spent on each activity as: (duration of activity ÷ total observation time) × 100.



Figure 2. South Philippine Hawk-Eagle (*Nisaetus pinsheri*) nestling during the late nestling stage (21 July 2023). (A) Nestling in defecating posture; (B) wing-flapping exercise; (C) standing position. Photo by R. L. Taraya.

We identified prey items delivered to the nest to the lowest taxonomic level possible, and described them in terms of whether they were brought whole or in parts. We included only prey identified to species level in biomass calculations, whereas unidentified rodents and bats were excluded. We derived some biomass estimates from local species-specific weight data: E. A. Lastica-Ternura provided the mean weight of an adult domestic chicken (*Gallus gallus domesticus*), while K. J. E. Suetos and F. P. O. Pagbilao provided weights for the green crested lizard (*Bronchocela cristatella*). We used published sources for Philippine Bush-Hen (*Amaurornis olivacea*; Taylor 2020) and Bagobo rat (*Bullimus bagobus*; Heaney et al. 2010). The green crested lizard, Philippine Bush-Hen, and Bagobo rat were always delivered whole as prey, so we used their reported average weights directly as biomass values. Although domestic chickens were delivered whole, they were visibly smaller than full-grown adults. Because Lastica-Ternura's reference weight (~ 2 kg) applies to adult domestic chickens, we conservatively estimated each of the five chickens brought to the nest at 0.50 kg, reflecting their smaller size. We calculated the percentage frequency of each prey item as (frequency of a specific prey item $\div$ total number of prey items) $\times 100$. We calculated the biomass of

each prey item and its relative contribution to the total prey biomass as (individual biomass $\div$ total biomass) $\times 100$.

Among the nestling behaviors we recorded, we most frequently observed head movement (24%), typically lasting from a few seconds to several minutes. We observed branching behavior, which began on 2 August 2023, as the second most frequent activity (14%; Table 1). This gradually progressed into short flights during the pre-fledging period, becoming increasingly frequent and aligning with flying exercises (31%, $n = 52$) involving brief flights between nearby branches or adjacent perches. The nestling took its first flight on 12 August 2023. After fledging, the juvenile remained near the nesting area for at least 2 wk, frequently returning to or staying within visual range of the nest tree. By 25 August 2023, we saw the juvenile flying to trees approximately 70–80 m away, vocalizing loudly, and no longer returning to the nest. Other common behaviors included standing and lying down, both in the nest and along its edge. Less frequent but notable activities, such as object play, preening, stretching, and self-feeding, accounted for 35% of the nestling's activity budget.

Throughout the observation period, we observed the female more frequently near the nest than the male, often perching on the nest tree itself or on

Table 1. The developmental milestones of the nestling South Philippine Hawk-Eagle (*Nisaetus pinskeri*).

Date	Developmental Milestones
21 July 2023	Standing upright; feeding on its own; flapping exercise
2 August 2023	Practicing flights from nest to branch or vice versa; branching
12 August 2023	First flight off the nest
25 August 2023	Flying to trees approximately 70–80 m from the nest with loud vocalizations; no nest attendance

nearby trees typically within ~20 m of the nest and oriented toward the nestling. The adult pair continued to perform mutual soaring displays above the nest area, a behavior typically associated with courtship and pair bonding. These aerial displays occurred even during the nestling-rearing period, suggesting that the pair maintained their social connection through ongoing mating rituals (Ibañez et al. 2003, Philippine Eagle Foundation 2009). As the nestling matured, we saw both adults leaving the nest area more often. Although we did not record direct nest defense events, we did see that the female was regularly targeted by mobbing Philippine Jungle Crows (*Corvus philippinus*). These disturbances were often intense and prolonged. For example, on 13 August 2023, the female was harassed by a group of crows for several minutes, and again on 22 August, she was first mobbed by crows and then attacked by a Peregrine Falcon (*Falco peregrinus*). The brief aerial fight between the two marked the only recorded interaction with another raptor during the monitoring period and may have been a territorial dispute or competition for aerial space. Despite these encounters, the adult pair remained attentive, displaying subtle but consistent behaviors such as proximity, vigilance, and social cues.

We documented 23 prey items delivered to the nest, including birds, mammals, and reptiles (Table 2).

Mammals made up 70% ($n = 16$) of all prey items. The second most common prey class was birds, including the domestic chicken and Philippine Bush-Hen. In terms of biomass contribution, birds accounted for the largest share (78%), driven primarily by the high biomass of domestic chickens. However, reliance on domestic chickens poses a serious conservation concern. Retaliatory shootings and trapping by farmers in response to poultry predation have become a major threat to the South Philippine Hawk-Eagle, as evidenced by numerous rescue cases involving injured or captured individuals (T. L. P. Senarillos unpubl. data). Additionally, such interactions may facilitate the spillover of avian diseases from domestic animals to wild raptors (Ayala et al. 2020). Addressing this growing human-wildlife conflict is critical and warrants further study and targeted mitigation efforts.

Parental roles in South Philippine Hawk-Eagle apparently mirror those of North Philippine Hawk-Eagle (*N. philippensis*), with the female performing more in-nest duties than the male (Gan et al. 2022). The male delivered nine prey items and the female seven. The male typically left the nest immediately after delivering food, while the female remained and fed the nestling. She was observed plucking the feathers or fur from the prey and tearing the meat into pieces before feeding it to the nestling. This feeding gradually decreased as the nestling matured. Each time the male delivered food, the nestling exhibited mantling behavior, spreading its wings and hunching over the prey, partially covering it with its wings and tail. We did not observe this behavior in the presence of the female.

CONCLUSIONS

Although our observations captured only the last part of one pair of South Philippine Hawk-Eagles' nesting behavior, they are useful, as no published information currently exists on the species' breeding behavior. We recommend the following future research

Table 2. Summary of the prey items delivered by the adult South Philippine Hawk-Eagles (*Nisaetus pinskeri*).

Class	Species	Number (% of Total)	Mass of Each (kg)	Total Biomass (kg)	Biomass Contribution (% of Total)
Birds	Domestic chicken (<i>Gallus gallus domesticus</i>)	5 (21.7)	0.50	2.50	69.0
	Philippine Bush-Hen (<i>Amaurornis olivacea</i>)	1 (4.3)	0.31	0.31	8.6
Mammals	Unidentified rats	11 (47.8)	UNK	UNK	UNK
	Bagobo rat (<i>Bullimus bagobus</i>)	2 (8.7)	0.40	0.80	22.1
	Unidentified bats	3 (13)	UNK	UNK	UNK
Reptiles	Green crested lizard (<i>Bronchocela cristatella</i>)	1 (4.3)	0.014	0.014	0.4
Total		23 (100)	NA	3.624	100

directions: (1) document egg-laying and hatching dates, (2) locate additional nests to allow for more detailed documentation of prey, behavior, and breeding success, (3) collect detailed data on nest and nest-site characterization to understand habitat preferences, and (4) conduct telemetry studies to determine juvenile dispersal patterns and the home range use by breeding adults. Investing in these research priorities will gradually build our understanding of the natural history of this globally threatened yet poorly studied Philippine raptor, and help shape more effective conservation strategies.

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